

HYDROFARMS AGRI SOLUTIONS

CONCRETE FLOORS, INTERLOCK PAVING, EQUIPMENT BASES AND FINISHES

Activity-Specific Construction Method Statement

Document field	Controlled value
Document code	HF-GH01-MST-CIV-002
Revision	00 - Draft for coordination
Project	One-Feddan NFT Lettuce Greenhouse Project
Location	Alexandria, Egypt
Prepared by	Hydrofarms Agri Solutions
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Parent methods	HF-GH01-MST-GEN-002, HF-GH01-MST-CIV-001 and HF-GH01-MST-CIV-004

ECONOMICAL HYBRID FLOOR SYSTEM
Concrete is retained at hygienic, central-circulation, service and equipment areas. Removable interlock is used in secondary production aisles. GROUNDCOVER 100 WHITE LINED is installed beneath tables on a smooth prepared substrate.

Document Control

Approval and responsibility

Role	Primary responsibility
Project manager	Resources, programme, subcontractor coordination and approval status.
Construction manager	Pour/paving sequence, work-front interfaces and daily release.
Civil engineer	Subgrade, base, formwork, concrete, pavers, joints and equipment plinths.
MEP engineer	Drains, sleeves, equipment dimensions, plinth openings and housekeeping interfaces.
Structural engineer	Equipment-base reinforcement and any structural foundation detail.
Surveyor	Floor zones, levels, falls, joints, drain levels and as-built survey.
QA/QC engineer	ITP, material tests, density, concrete, finish and handover records.
HSE manager	Concrete/paving hazards, access, segregation and chemical controls.
Client/consultant	Hold/witness inspection and technical acceptance.

Revision history

Rev.	Date	Purpose	Description
00	15 Aug 2026	Draft for coordination	Initial project-specific English issue

1. Purpose

This method defines the construction and acceptance of production/service greenhouse floors, interlock paving, floor drains, groundcover substrate and equipment housekeeping bases. It ensures that buried services remain accessible and tested, hygiene areas drain correctly and the economical floor mix is built without quantity duplication.

2. Scope and Package Boundary

The civil flooring package includes the following controlled works:

- Subgrade preparation, local replacement, granular base, polyethylene membrane and floor-edge restraint.
- Production-greenhouse concrete aisles/margins, secondary-aisle interlock and prepared substrate beneath groundcover.
- Full concrete flooring in the 9 x 36 m service greenhouse and 4 x 4 m sterilisation room.
- Drain bodies, trenches/recesses, formed falls, movement/construction joints and surface transitions.
- Concrete housekeeping plinths for pumps, chillers, WTP skids and other equipment where shown on AFC drawings.
- Protection, curing, cleaning, testing and as-built handover before installation of tables or sensitive equipment.

NO DUPLICATION
Main greenhouse steel foundations and anchor bolts belong to the structural-steel package. Buried pipes belong to HF-GH01-MST-CIV-004. Groundcover material procurement belongs to the covering package; this civil method includes its prepared substrate and coordinated installation only.

3. Governing Information

Priority	Controlled information
1	AFC floor zoning, levels, falls, joint layout, drains and equipment-base drawings
2	Accepted buried-utility tests and as-built survey under HF-GH01-MST-CIV-004
3	Approved geotechnical report and subgrade/base material recommendations
4	Approved concrete mix, reinforcement, paver, membrane and joint material submittals
5	Equipment supplier base-load, footprint, anchor and maintenance-clearance data
6	Approved ITP, pour plans, trial panels and curing/protection procedures

PRE-CLOSURE HOLD

No permanent concrete, paver base or groundcover substrate shall conceal a utility route until testing, invert/coordinate survey, photographs and concealed-work release are complete.

4. Approved Floor Zoning

Area	Approved system	Area	Principal build-up
Production main/service aisles	Reinforced slab-on-ground	803.04 m ²	100 mm C25 concrete
Production secondary aisles	Concrete interlock pavers	1,344.00 m ²	60 mm paver + 30-40 mm sand + 150 mm base
Below 84 tables	GROUNDCOVER 100 WHITE LINED	2,388.96 m ²	30 mm fine-sand levelling on prepared substrate
Service greenhouse	Full reinforced concrete floor	324.00 m ²	120 mm C25 + membrane + 150 mm base
Sterilisation room	Full reinforced concrete floor	16.00 m ²	100 mm C25 + membrane + 150 mm base

The production greenhouse footprint is 126 x 36 m = 4,536 m². The 803.04 m² production-concrete quantity already includes 299.04 m² of 100 mm concrete margins extending below table edges; that margin shall not be added again.

Floor geometry controls

- Maintain the 2.5 m central production corridor and the approved fan/pad service aisles as continuous concrete routes.
- Maintain 0.75 m clear operating aisles between NFT tables. Paver/concrete interfaces shall not reduce that clear width.
- Extend concrete 100 mm beneath each table perimeter as the adopted durable edge; the remaining table footprint receives groundcover.
- Service-greenhouse flooring is fully concrete because it serves germination, packing, washing and hygienic circulation.

5. Material and Quantity Schedule

Item	Installed quantity	Procurement quantity	Minimum specification
Subgrade preparation	4,876 m ²	4,876 m ²	Grading and density testing
Compacted granular base	403 m ³	403 m ³ compacted	Approved graded crushed aggregate, 95% MDD
Polyethylene membrane	1,143.04 m ² concrete footprint	1,258 m ²	250 micron with sealed laps
Production floor concrete	80.30 m ³	82.71 m ³	C25, 100 mm, includes 3% procurement waste
Service/sterilisation concrete	40.48 m ³	41.70 m ³	C25, 100/120 mm, includes 3% waste
Production welded mesh	803.04 m ²	884 m ²	6 mm @ 150 x 150 mm including laps
Service/sterilisation mesh	340 m ²	374 m ²	8 mm @ 200 x 200 mm including laps
60 mm interlock pavers	1,344 m ²	1,412 m ²	At least 40 MPa, 5% cutting allowance
Paver bedding sand	53.76 m ³	57 m ³	Washed sharp sand, 30-40 mm compacted bed
Paver jointing sand	5.4 t	6.0 t	Dry/polymeric compatible sand
Paver edge restraints	Measured work	1,800 m pricing allowance	At aisles, openings, drains and transitions
Groundcover fine levelling	71.67 m ³	76 m ³	30 mm clean fine sand, no sharp stones
Groundcover material	2,388.96 m ²	2,510 m ²	Procured once in covering package
Floor drain bodies	19 pc	19 pc	15 production + 4 service/sterilisation

6. Preconditions

- All buried feed/return/drainage routes are tested, flushed, surveyed and released before base-course closure.
- Steel-column foundations/anchors and fan/pad frames are either complete or protected by an approved construction sequence.
- Drain invert and frame levels, equipment plinths, door thresholds and final external levels are coordinated.
- Approved trial panels establish concrete finish, paver pattern, joint filling, surface tolerance and cleanability.

- Concrete supply, reinforcement, compaction plant, curing materials and weather controls are available.
- Work zones permit movement without contaminating accepted groundcover areas or damaging exposed utilities.

7. Detailed Execution Method

7.1 Survey and formation preparation

- Transfer floor zones, slab/paver edges, table footprints, drain centres, joint lines and equipment-base coordinates from approved control points.
- Proof-roll or inspect subgrade, remove soft/organic/contaminated material and replace it by the approved earthwork method.
- Trim and compact formation to level and achieve minimum 95% laboratory maximum dry density in hardscape areas.
- Protect valve boxes, cleanouts, sleeves and survey points so final finishes remain accessible and correctly aligned.

7.2 Granular base and membrane

- Place approved graded base uniformly in controlled lifts, condition moisture and compact to minimum 95% MDD.
- Survey finished base levels and falls; correct low areas by reworking the layer, not by loose sand patches under concrete.
- Place 250 micron polyethylene membrane beneath concrete on a smooth surface; overlap, tape and seal penetrations without tears.
- Protect the membrane during reinforcement and concreting. Patch damage with compatible material before the pre-pour inspection.

7.3 Formwork, reinforcement and embedded items

- Set forms to approved edges, levels, falls, construction joints and isolation joints around columns, walls, plinths and chambers.
- Support mesh at the specified cover using stable non-corroding spacers; mesh lying on the membrane/subbase is unacceptable.
- Provide 6 mm mesh at 150 x 150 mm in production concrete and 8 mm mesh at 200 x 200 mm in service/sterilisation floors.
- Install drain bodies, sleeves, trench frames, anchor templates and cast-in items; secure them against flotation or movement.

7.4 Production concrete floor

- Place C25 concrete at 100 mm nominal thickness by an approved panel sequence that maintains access and joint continuity.
- Compact without displacing mesh, drains or sleeves; screed to the approved 0.5% drainage falls and finish non-slip and washable.
- Form the 100 mm concrete intrusion beneath table edges accurately; protect the adjoining fine-sand/groundcover zone from concrete spillage.
- Finish transitions flush with pavers and access boxes; abrupt lips that obstruct carts or cleaning are prohibited.

7.5 Service greenhouse and sterilisation floors

- Place 120 mm C25 floor in the 324 m² service greenhouse and 100 mm C25 floor in the 16 m² sterilisation room.
- Provide a dense smooth non-slip finish with 0.5-1.0% falls toward the four service/sterilisation drains.
- Form hygienic coved epoxy or compatible cementitious skirting at wall/floor junctions in wash areas after concrete moisture criteria are achieved.
- Coordinate door thresholds, central anti-virus corridor, packaging zone and northern germination-zone circulation without ponding.

7.6 Interlock paving

- Install concrete edge restraints and allow them to reach adequate strength before laying pavers.
- Spread 30-40 mm washed sharp bedding sand over the accepted base without trafficking the screeded bed.
- Lay 60 mm pavers in the approved pattern, maintaining line, level, drainage and tight uniform joints; cut with mechanical saws.
- Compact with a suitable protected plate, fill joints with approved dry/polymeric sand and repeat until joints are full and stable.
- Keep paver surfaces flush at concrete, drains, inspection boxes and table access points; rocking, broken or stained units shall be replaced.

7.7 Groundcover substrate and installation interface

- Complete heavy work, wet testing and overhead contamination-risk activities before final groundcover installation.
- Place 30 mm clean fine levelling sand free from sharp stones; finish smooth, stable and without hollows that retain wash water.

- Lay GROUNDCOVER 100 WHITE LINED without wrinkles, punctures or exposed sharp edges and coordinate laps/fixings with the covering supplier.
- The installed target is 2,388.96 m²; procurement quantity 2,510 m² is counted once under the covering package.

7.8 Equipment housekeeping bases

- Confirm actual equipment footprint, operating weight, maintenance removal path, anchor template and pipe/cable approach before casting.
- Construct equipment plinths in the concrete grade, reinforcement, dimensions and elevation shown on AFC structural/equipment drawings.
- Provide isolation joint from surrounding floor where detailed, non-shrink grout beneath steel baseplates and chamfered durable edges.
- Do not use floor screed or loose shims as equipment bearing. Protect anchors and verify projection, spacing and verticality before handover.

7.9 Joints, curing and protection

- Form construction, contraction and isolation joints to the approved layout; align joints with geometry changes and avoid random re-entrant cracking.
- Saw-cut contraction joints at the approved time/depth before uncontrolled cracking develops and seal where hygiene or water control requires it.
- Cure concrete continuously by the approved method and restrict traffic until the required strength is achieved.
- Protect finished floors/pavers from oil, welding, impact, concentrated loads and construction traffic; repair damage by an approved method.

8. Drainage and Surface Tolerances

Characteristic	Acceptance criterion
Production concrete falls	0.5% toward approved drainage points; no reverse fall
Service/sterilisation falls	0.5-1.0% toward trapped drains
Paver/concrete deviation	Maximum 5 mm under a 2 m straightedge
Ponding	No standing water after functional washdown/drain test
Floor drain count	15 production corridor drains + 4 service/sterilisation drains
Drain cover	Cleanable, load-suitable and insect-protected; stainless or engineering plastic in hygiene areas
Transitions	Flush, stable and safe for personnel/manual harvest carts
Groundcover substrate	Smooth, stable and free from sharp stones, debris and local water traps

9. Concrete Quality Control

- Verify approved mix designation, delivery ticket, time, slump/workability and temperature for every load.
- Take strength specimens at the approved frequency and identify them by pour, location and batch.
- Reject concrete outside approved time/workability limits; uncontrolled site water addition is prohibited.
- Check thickness by controlled level pins and agreed cores where required; assess cracks, laitance, honeycombing and finish defects.
- Complete a representative trial panel before bulk work and retain it as the finish/tolerance benchmark.

10. Inspection and Test Plan

No.	Inspection stage	Method	Type	Responsibility	Record
01	AFC zoning, levels, joints and interfaces	Document review	H	Civil/MEP/QA-QC	Work-front release
02	Buried utility tests/as-built acceptance	Record/survey review	H	MEP/Survey/Consultant	Closure release
03	Subgrade condition and density	Survey/field test	H	Civil/QA-QC/Lab	Formation report
04	Granular base level and density	Survey/field test	H	Civil/QA-QC/Lab	Base report
05	Membrane, forms, mesh, drains and embeds	Pre-pour inspection	H	Civil/MEP/Consultant	Pour card
06	Concrete delivery and placement	Site/lab testing	W	Civil/QA-QC/Lab	Concrete report
07	Finish, joints, curing and protection	Visual/measurement	W	Civil/QA-QC	Floor inspection
08	Paver base, restraints and sand bed	Level/density/visual	H	Civil/QA-QC	Paving release
09	Pavers, joints and surface tolerance	Straightedge/visual	W	Civil/Consultant	Paver report
10	Drainage/washdown test	Functional test	H	Civil/MEP/Consultant	Drainage report
11	Equipment plinths and anchors	Survey/dimensional	H	Civil/MEP/Consultant	Plinth release
12	Groundcover substrate/install interface	Visual/level	W	Civil/Covering/QA-QC	Interface release
13	Final as-built and handover	Survey/record review	R	QA-QC/Client	Turnover dossier

Intervention codes: H = Hold, W = Witness, S = Surveillance, R = Record review.

11. Hold Points

- HP-FL-01: Buried utilities tested, photographed and surveyed before floor/base closure.
- HP-FL-02: Subgrade and granular base density/levels accepted before membrane or bedding sand.
- HP-FL-03: Formwork, joints, reinforcement, drains, sleeves and embeds accepted before concrete.
- HP-FL-04: Trial concrete and paver panels accepted before bulk installation.
- HP-FL-05: Equipment plinth geometry and anchor templates accepted before concrete.
- HP-FL-06: Drainage falls and washdown test accepted before final finishes/groundcover.
- HP-FL-07: Final surface, joints, cleanliness and records accepted before tables/equipment are released for installation.

12. HSE and Environmental Controls

Hazard	Mandatory control	Residual risk
Concrete plant/vehicles	Segregated route, banksman, lighting and exclusion zone	Medium
Wet concrete/cement	Gloves, eye/skin protection, wash facilities and SDS controls	Low
Mesh/cuts/impalement	Capped bars, gloves, safe storage and controlled cutting	Low
Compaction vibration/noise	Suitable equipment, hearing protection and exposure control	Low
Saw cutting/dust	Guarded wet-cut equipment, respiratory/eye protection and slurry collection	Low
Slips and access	Barricades, defined walkways, housekeeping and adequate curing-zone control	Low
Concrete washout	Dedicated lined washout; no discharge to drains, tanks or soil	Low

13. Records and Handover

- Floor-zone survey, utility closure release, material approvals and subgrade/base density reports.
- Concrete delivery, slump, specimen, strength, pour-card, curing and defect-repair records.
- Paver/trial-panel records, drain tests, groundcover interface release and final-cleaning photographs.
- Equipment plinth/anchor survey, as-built floor/joint/drain drawings, NCR closeout and signed area turnover.